
Regularization and Its Effects on Spurious Feature Learning

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Abstract

Modern neural networks often achieve high accuracy by relying on spurious features, shortcuts that correlate with the label in training but fail to generalize. While prior work (Izmailov et al., 2022) has shown that core features are still learned even when spurious shortcuts are present, less is understood about how regularization reshapes the internal trajectory of feature emergence across depth and training time. In this work, we extend the linear probing framework to study how L1, L2, and dropout regularization alter where and when spurious versus core features become linearly accessible in a ResNet-18. Using Colored MNIST and Waterbirds as controlled testbeds, we attach probes at six depths and track per-layer accuracy across epochs. We find that regularization does not simply suppress spurious features uniformly, instead each method reshapes the depth-wise trajectory differently. Moderate regularization restricts spurious feature propagation into deeper layers without harming core feature learning, while aggressive regularization destabilizes both. Our results suggest that the location of spurious encoding, not just its presence, is a useful lens for understanding regularization’s effect on shortcut learning.

1. Introduction

Deep neural networks perform very well on many tasks, but they often rely on spurious features—patterns in the data that are correlated with the label but are not actually meaningful. For example, a model might use the background color of an image instead of the object itself to make predictions. This can lead to poor performance when the data changes, especially for rare or minority cases.

Previous work by Izmailov et al shows that even when models rely on these shortcuts, they may still learn the correct core features internally. The main difference is how these features are used in the final layer. However, this work only looks at the final trained model and does not explain how these features develop during training or where they appear inside the network.

In this project, we study how core and spurious features are learned over time and across different layers of a neural network. Instead of only looking at the final output, we examine what the model is learning at different stages of training. We ask questions like: When does the model first learn the spurious feature? When does it learn the core feature? And at what point does the core feature become more important than the spurious one?

We are also interested in how regularization affects this process. Methods like L1/L2 regularization and dropout are often used to make models more robust. However, it is not clear whether they actually change what the model learns, or just change how the final layer uses the learned features. We study whether regularization delays or reduces the learning of spurious features and helps the model focus more on core features.

To do this, we train a ResNet-18 model and use linear probes to analyze its internal layers. At different points during training, we extract activations from each layer and train simple classifiers to predict both the core feature (e.g., digit label) and the spurious feature (e.g., color). This allows us to see which features are present at each layer and time step.

We run experiments on the Colored MNIST dataset, where color is intentionally correlated with the label, and extend our analysis to the Waterbirds dataset, where background acts as a spurious feature. By tracking how features appear and change over time, we aim to better understand how neural networks learn shortcuts and how regularization can help reduce this behavior.

2. Methods

2.1. Data Collection

We experiment with two different datasets: the Colored MNIST dataset from (Arjovsky et al., 2019), and the Waterbirds dataset created by (Sagawa et al., 2020). Both are specifically designed to introduce spurious features that models can rely on to achieve high performance on binary classification problems.

2.1.1. COLORED MNIST

This dataset is based on MNIST (LeCun et al., 1998), but is formulated as a binary classification problem where digits 0, 1, 2, 3, 4 have label 0 and digits 5, 6, 7, 8, 9 have label 1. Each image is colored either red or green to introduce a spurious feature. Following the procedure in (?), the color is determined by the label: one color is assigned to each class, but this assignment is flipped with probability $1 - s$, where s is the desired spurious correlation. We use $s = 0.9$, so 90% of images have the color that matches their label and 10% have the opposite color. A model that relies on color as a shortcut can therefore achieve up to 90% training accuracy without learning digit shape at all, making this dataset a useful testbed for studying how and where spurious features are encoded across layers.

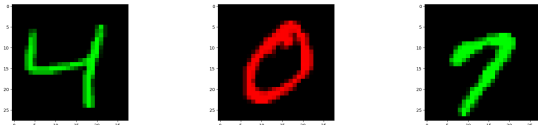


Figure 1. Sample images from the Colored MNIST dataset.

2.1.2. WATERBIRDS

The Waterbirds dataset, introduced by (Sagawa et al., 2020), is constructed from the CUB bird dataset (Wah et al., 2011) and the Places dataset (Zhou et al., 2017). Each bird is labeled as either a waterbird or a landbird, and is placed onto either a water or land background using pixel-level segmentation masks. The background serves as the spurious feature: in the training set, 95% of waterbirds appear on water backgrounds and 95% of landbirds appear on land backgrounds. As with Colored MNIST, a model can exploit this correlation and classify based on background alone, reaching high training accuracy without learning to identify the bird itself. We use this dataset as provided, without any modifications to its construction.



Figure 2. Sample images from the Waterbirds dataset.

2.2. Linear Probes

To measure how much information about each feature is present at different depths of the network, we attach linear

probes (Alain & Bengio, 2017) to the intermediate activations of the ResNet18 (He et al., 2016) following the forward hooks approach detailed in (Bhaskar, 2021). A linear probe is a single linear classifier trained on the frozen activations of a given layer. If a probe achieves high accuracy on a particular feature, that feature is linearly accessible in that layer’s representation. By training separate probes for the core feature (digit class or bird type) and the spurious feature (color or background), we can track where and when the network encodes each one during training.

We attach probes after six points in the ResNet18: the initial ReLU, the outputs of `layer1` through `layer4`, and the global average pooling layer. At each point, we register a forward hook that captures the activation tensor during the forward pass. The probe takes this activation, applies adaptive average pooling to collapse the spatial dimensions, and feeds the result into a linear probe that outputs a 2-class prediction.

2.3. Training

We train a ResNet18 trained from scratch for both datasets. For Colored MNIST, we modify the first convolutional layer to use a 3×3 kernel with stride 1 and padding 1, and replace the max pooling layer with an identity operation, to accommodate the 28×28 input size. For Waterbirds, we use the standard ResNet18 architecture without modification. In both cases, the final fully connected layer is replaced with a 2-class output.

All models are trained for 10 epochs using SGD with a learning rate of 0.01, momentum of 0.9, and a batch size of 32. The probes are trained jointly during each training step: after the model performs its forward and backward pass, the probes compute their own loss on the detached activations and update independently via a separate SGD optimizer with a learning rate of 0.01. Probe accuracy for each layer and feature type is logged at the end of every epoch.

3. Results

For each regularization method, we train three models at increasing regularization strengths and examine how the probe accuracy heatmaps change relative to an unregularized baseline. We report overall model accuracy, precision, and recall alongside the heatmaps.

To visualize the results of the linear probes, we plot heatmaps where the x-axis represents the training epoch and the y-axis represents the layer depth, from the initial ReLU through the global average pool. Each cell shows the probe accuracy for a given layer at a given epoch, colored on a red-to-green scale from 0 to 1. We generate one heatmap for the core feature probe (digit or bird type) and one for the

spurious feature probe (color or background).

3.1. Colored MNIST

3.1.1. BASELINE (NO REGULARIZATION)

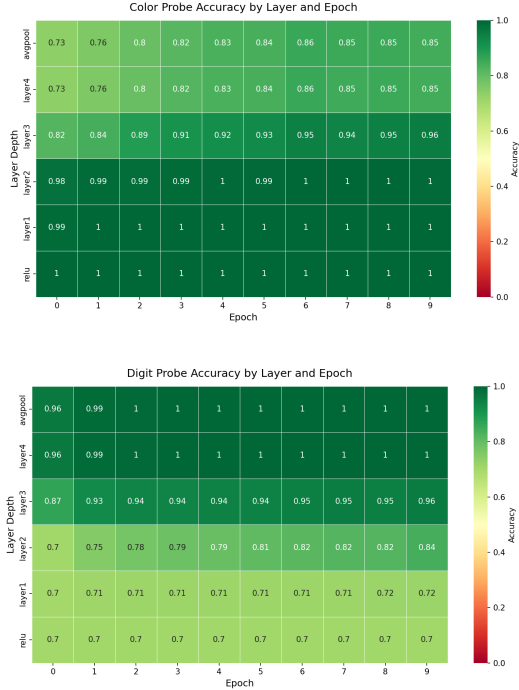


Figure 3. Heatmaps for the unregularized baseline model.

Without any regularization, the model learns both features quickly. Color probe accuracy saturates in the early layers within the first epoch and remains high throughout training. Over time, color also becomes increasingly readable in the deeper layers, spreading from the early layers into layer3, layer4, and avgpool. Digit probe accuracy builds progressively with depth, reaching near-perfect accuracy in the later layers. The model achieves 99.29% accuracy, 99.69% precision, and 98.85% recall. This baseline establishes the default behavior against which we compare all regularized models: early and persistent encoding of the spurious feature, with gradual spread into deeper layers over training.

3.1.2. L1 REGULARIZATION

Figure 4 shows the accuracies of the linear probes under different levels of L1 regularization.

At $\lambda = 1e-5$, the model quickly learns strong representations for both features. Color is encoded prominently in the early layers (ReLU and layer1) from the very first epoch, with near-perfect probe accuracy (0.99). Digit features are weaker in the early layers but improve substantially in the

deeper layers (layer3, layer4, avgpool), reaching 0.98 by the final epoch. As training progresses, color probe accuracy also rises in the deeper layers, suggesting that the spurious feature gradually spreads into later representations. The model achieves 99.40% accuracy, 99.59% precision, and 99.18% recall.

At $\lambda = 1e-3$, color features are still learned early but are less dominant than in the weaker setting. Early layers (ReLU, layer1) initially show very high color probe accuracy (0.99, 0.97), but this decays over training epochs rather than spreading into deeper layers, suggesting that moderate L1 constrains the persistence of the spurious feature. Digit features improve steadily with depth, reaching 0.98 in the later layers by the final epoch. The result is a more balanced representation: spurious features are present but do not dominate, and core features are well-captured in deeper layers. The model achieves 97.79% accuracy, 98.35% precision, and 97.08% recall.

At $\lambda = 1e-2$, the behavior changes substantially. Color probe accuracy is weak and inconsistent across all layers, never exceeding 0.81, and digit probe accuracy remains low throughout training, plateauing near chance level in the early layers and only reaching 0.83 in the deepest layers. The heavy sparsity penalty suppresses both the spurious and core feature representations, causing the model to underfit severely. The model achieves only 54.26% accuracy, 94.43% precision, and 6.27% recall, indicating that it has largely collapsed to predicting a single class.

Across the three settings, increasing L1 strength progressively constrains feature learning across all layers. Unlike L2, where moderate regularization selectively prevents spurious features from leaking into deeper layers, L1’s sparsity pressure suppresses features more uniformly. Weak L1 ($1e-5$) leads to strong shortcut reliance as color spreads across the network; moderate L1 ($1e-3$) achieves the best trade-off by limiting spurious feature persistence without harming core feature learning; and strong L1 ($1e-2$) causes underfitting by indiscriminately suppressing both spurious and core representations.

3.1.3. L2 REGULARIZATION

Figure 5 shows the accuracies of the linear probes under different levels of L2 regularization. At $\lambda = 1e-5$, the model learns strong representations for both features within the first epoch. Color is encoded in the early layers (ReLU through layer2) and digit shape is encoded in the later layers (layer3, layer4, avgpool). As training continues, color probe accuracy increases in the deeper layers as well, suggesting that the spurious feature spreads into the later representations over time. The model achieves 99.27% accuracy, 99.24% precision, and 99.26% recall.

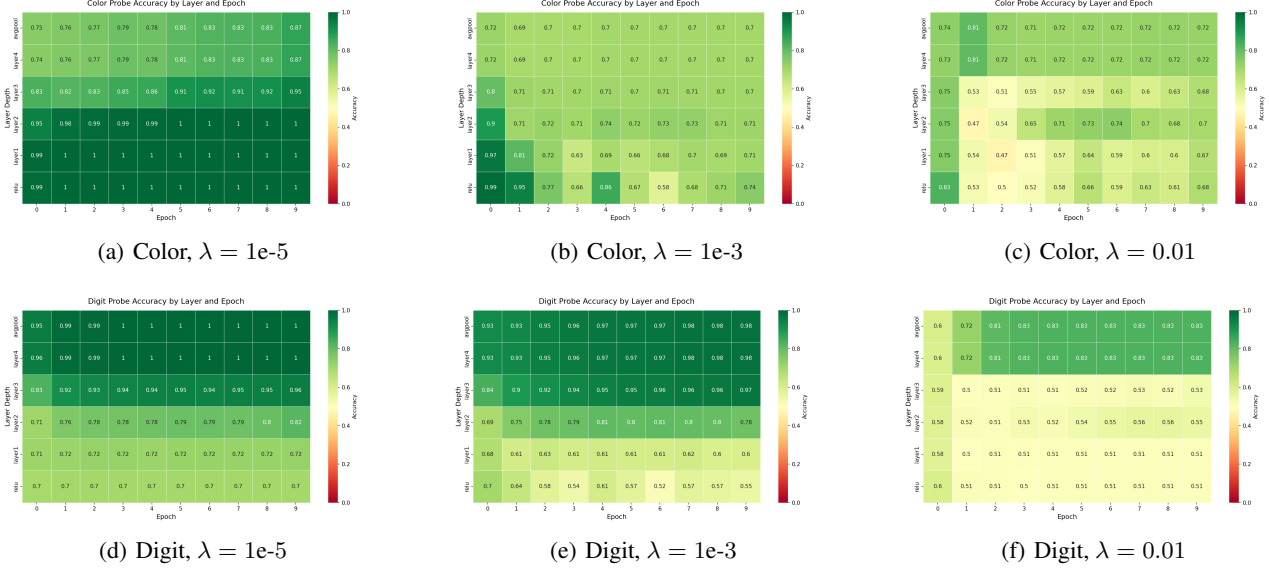


Figure 4. L1 regularization probe accuracies on Colored MNIST. Top row: color (spurious) probe. Bottom row: digit (core) probe. Regularization strength increases left to right.

At $\lambda = 1e-3$, the overall structure is similar: color in early layers, digits in later layers. However, unlike the weaker setting, color does not spread into the deeper layers over the course of training. It remains fixed in the early layers throughout. The model achieves 99.20% accuracy, 99.36% precision, and 98.99% recall.

At $\lambda = 0.1$, the behavior changes substantially. Color is no longer well-represented in the early layers and only becomes barely detectable in the final layers during the last epoch. Digit accuracy is initially strong but degrades in later epochs, indicating that the heavy weight penalty destabilizes learned representations over time. The model achieves 55.56% accuracy, 52.37% precision, and 94.82% recall.

Across the three settings, increasing L2 strength progressively constrains the spread of features across layers. Moderate L2 prevents the spurious feature from leaking into deeper layers without harming core feature learning, while aggressive L2 suppresses both features and causes underfitting.

3.1.4. DROPOUT

Figure 5 shows linear probe accuracies under three dropout rates: 0.1, 0.3, and 0.5. Across all settings, the model achieves strong overall classification performance (99.3% accuracy), but the internal feature trajectories differ meaningfully.

At $p=0.1$, the model behaves similarly to the unregu-

larized baseline. Color is fully encoded in the early layers and gradually seeps into the deeper layers as training progresses. Digit accuracy builds steadily with depth. This light dropout setting does not visibly alter the spread of the spurious feature across the network.

At $p=0.3$, the spurious feature is more clearly localized. Color remains strongly encoded in the early layers but its presence in deeper layers is reduced compared to $p = 0.1$. The crossover point, where digit accuracy overtakes color, occurs around layer2/layer3. This suggests that moderate dropout suppresses the propagation of spurious features into intermediate layers without harming core feature learning.

At $p=0.5$, dropout produces the strongest suppression of the spurious feature. Color still saturates in the very early layers, indicating that dropout does not affect the initial encoding of color. However, color accuracy in layer2 actually decreases over training, suggesting active suppression of spurious feature propagation rather than passive prevention. Layer3 and layer4 color probes stay flat and low, which is the strongest suppression observed across all regularization conditions. Digit accuracy still climbs normally through depth.

Unlike L2, which reduces feature spread by penalizing weight magnitudes globally, dropout appears to specifically interfere with the *propagation* of color through the residual blocks while leaving the initial encoding intact. This produces a more localized spurious representation and an earlier crossover between core and spurious features.

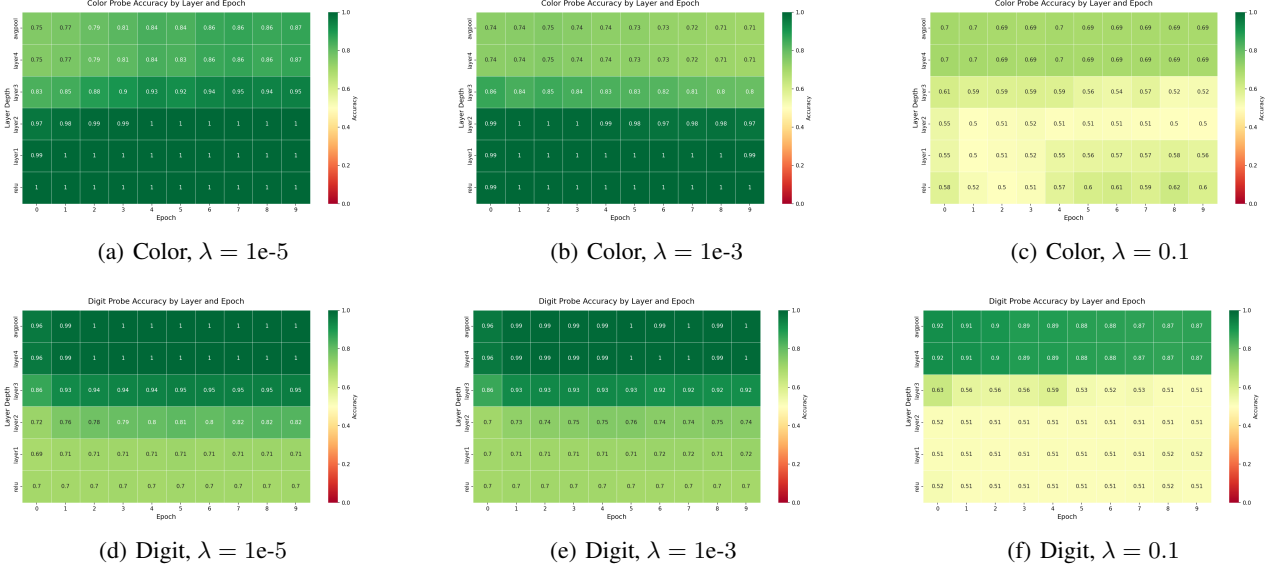


Figure 5. L2 regularization probe accuracies on Colored MNIST. Top row: color (spurious) probe. Bottom row: digit (core) probe. Regularization strength increases left to right.

3.2. Waterbirds

Across all three regularization methods, the Waterbirds dataset exhibits qualitatively similar trends to Colored MNIST: moderate regularization constrains spurious feature propagation into deeper layers without degrading core feature learning, while aggressive regularization suppresses both. However, the effects are less pronounced, likely due to the greater visual complexity of the background feature relative to digit color. Full heatmaps and per-method analysis for Waterbirds are provided in Appendix A.

4. Process

The main engineering effort was adapting ResNet-18 to work with 28×28 MNIST inputs, since the standard architecture assumes 224×224 images. We replaced the first convolutional layer with a smaller kernel and removed max pooling to prevent the spatial dimensions from collapsing too early. We also built the probing pipeline, which involved registering forward hooks at six layers, pooling the extracted activations, and training separate linear classifiers on detached activations so that probe updates would not affect the main model’s gradients.

The $\lambda = 1e-2$ L1 result initially appeared contradictory: the model reported 94.43% precision alongside only 6.27% recall and 54.26% overall accuracy. At first glance, high precision suggested the model was still making meaningful predictions, but the near-zero recall revealed that it had almost entirely collapsed to predicting a single class. Understanding this required looking at the predictions directly

rather than relying on accuracy alone, and reinforced that precision in isolation can be misleading when the model’s prediction distribution is heavily skewed.

5. Contributions

5.0.1. SOPHIA MIRRASHIDI

Implemented the linear probes, training pipeline, dataset generation, and heatmap plotting code. Wrote the Methods and L2 Regularization sections of this report. Coordinated meeting times and contributed to group discussions throughout the project.

5.0.2. VARAD TAWDE

Implemented the dropout regularization experiments, training and evaluating models on both Colored MNIST and Waterbirds. Wrote the Abstract and Dropout Regularization sections of this report. Contributed to group discussions, code reviews, and meeting coordination throughout the project.

5.0.3. KIRAN KHANNA

Implemented L1 regularization experiments, including model training and evaluation on both the Colored MNIST and Waterbirds datasets. Authored the Introduction and L1 Regularization sections of the report, and actively contributed to group discussions throughout the project.

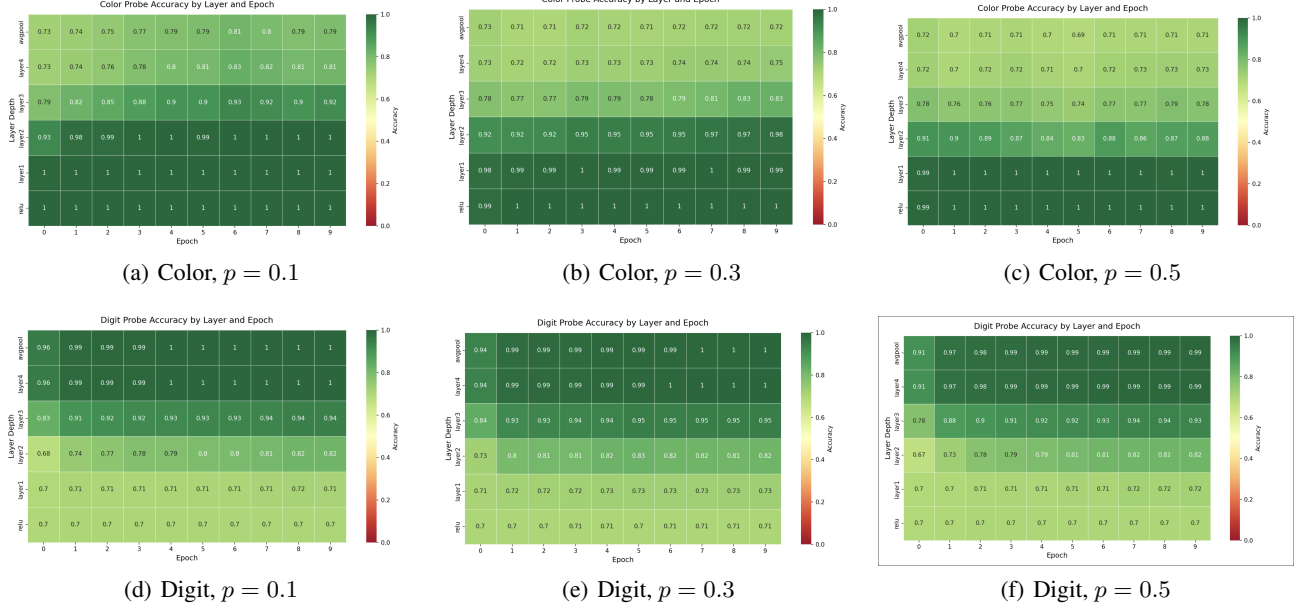


Figure 6. Dropout regularization probe accuracies on Colored MNIST. Top row: color (spurious) probe. Bottom row: digit (core) probe. Dropout strength increases left to right.

6. AI Usage

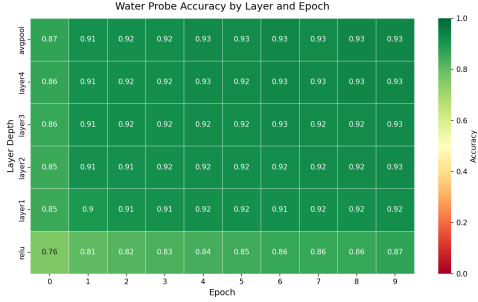
We used AI assistants (mainly Claude Opus 4.6) throughout this project. AI was used to assist with implementation of the dataset generation, training pipeline, and heatmap plotting code, as well as to help draft and edit sections of this report.

7. Github

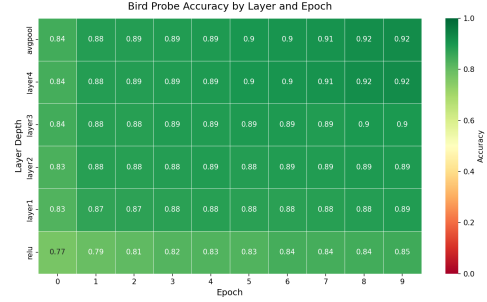
[Github](#).

References

- Alain, G. and Bengio, Y. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2017.
- Arjovsky, M., Bottou, L., Gulrajani, I., and Lopez-Paz, D. Invariant risk minimization. *arXiv preprint arXiv:1907.02893*, 2019.
- Bhaskar, N. Forward hooks in pytorch, 2021. URL <https://web.stanford.edu/~nanbhas/blog/forward-hooks-pytorch/>.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.
- Izmailov, P., Kirichenko, P., Gruver, N., and Wilson, A. G. On feature learning in the presence of spurious correlations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- LeCun, Y., Bottou, L., Bengio, Y., and Haffner, P. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- Sagawa, S., Koh, P. W., Hashimoto, T. B., and Liang, P. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. In *International Conference on Learning Representations*, 2020.
- Wah, C., Branson, S., Welinder, P., Perona, P., and Belongie, S. The caltech-ucsd birds-200-2011 dataset. Technical report, California Institute of Technology, 2011.
- Zhou, B., Laporta, A., Oliva, A., and Torralba, A. Places: A 10 million image database for scene recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(6):1452–1464, 2017.



(a) Water



(b) Bird

Figure 7. Unregularized Baseline model probe accuracies on Waterbirds. Top row: water (spurious) probe. Bottom row: bird (core) probe. Regularization strength increases left to right.

A. Waterbirds Heatmaps

A.1. Unregularized Baseline

Figure 7 shows the linear probe accuracies for the Waterbirds dataset without any regularization. Both the water (spurious) and bird (core) probes are detectable across all layers from the very first epoch, with no clear depth-based localization of either feature. The water probe is uniformly high across all layers, climbing from 0.85–0.87 at epoch 0 to 0.92–0.93 by epoch 9 in the deeper layers, with even the earliest layer (relu) reaching 0.87 by the final epoch. The bird probe follows a similar trajectory but remains consistently weaker at every layer and epoch, ranging from 0.77 at relu to 0.84 at avgpool at epoch 0, and improving to 0.85 and 0.92 respectively by epoch 9.

Unlike Colored MNIST, where the spurious color feature was concentrated in early layers, the background feature in Waterbirds is distributed across the entire network depth from the earliest stages of training. The persistent gap between the water and bird probes at every layer throughout training suggests that the model consistently prioritizes the spurious background over the core bird feature, a pattern consistent with shortcut learning driven by the dataset’s strong spurious correlation. The bird probe’s depth-dependent improvement — with deeper layers reaching 0.92 versus relu’s 0.85 — indicates that the core feature requires more compositional processing to extract, yet the spurious feature remains dominant regardless of depth.

A.2. L1 Regularization

Figure 8 shows the linear probe accuracies for the Waterbirds dataset under different levels of L1 regularization. As with Colored MNIST, the three settings reveal distinct regimes of feature learning, though the spurious and core features are less cleanly separated across layers than in the digit task.

At $\lambda = 1e-5$, both features are detectable across all layers from the start of training, but the water (spurious) probe dominates uniformly. Water probe accuracy is high across all layers from the first epoch, ranging from 0.87 at relu to 0.94 at avgpool by epoch 9, and improves steadily throughout training. The bird (core) probe is notably weaker and more depth-dependent: early layers (relu) plateau around 0.83, while deeper layers (layer4, avgpool) improve gradually to 0.94 by the final epoch. Unlike Colored MNIST, where features were more layer-localized, both features here are readable at every depth, but the spurious background feature maintains a consistent advantage across layers. This mirrors the weak L1 behavior seen in Colored MNIST: the model encodes the shortcut feature strongly and broadly, with core feature learning improving only gradually in the deeper layers.

At $\lambda = 1e-3$, bird probe accuracy is more uniformly distributed across layers, with all layers from layer1 through avgpool achieving around 0.88–0.94 by the final epoch. Compared to $1e-5$, the bird probe is slightly weaker in the deepest layers (peaking at 0.94 vs. 0.94, with less consistent improvement mid-training) but remains stable. The water probe, however, shows a notable change: rather than being uniformly high across all layers, it is concentrated in the deeper layers (layer4, avgpool reaching 0.85–0.86) while the early and middle layers (relu through layer3) plateau near 0.74, close to the majority-class baseline. This suggests that moderate L1 partially suppresses the spurious background feature in early layers, though it still leaks into the deeper representations.

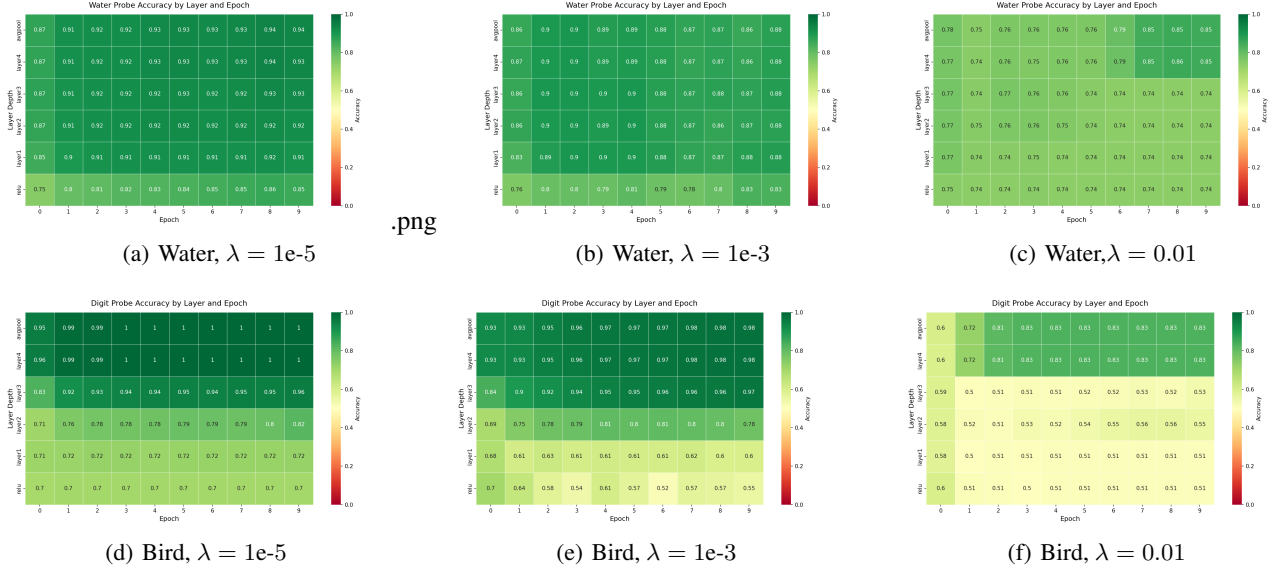


Figure 8. L1 regularization probe accuracies on Waterbirds. Top row: water (spurious) probe. Bottom row: bird (core) probe. Regularization strength increases left to right.

At $\lambda = 1e-2$, the pattern is more uniform and subdued than in Colored MNIST, but the effect of heavy regularization is still visible. The bird probe is moderate and relatively flat across all layers, with values clustering around 0.77–0.83 regardless of depth or epoch — showing little differentiation between early and late layers. The water probe is similarly constrained, hovering between 0.76 and 0.90 across layers with no clear depth-dependent structure. Unlike Colored MNIST, where strong L1 caused a near-total collapse in recall, the Waterbirds results do not show as dramatic a degradation, likely because the overall feature complexity is lower and neither feature achieved near-perfect probe accuracy even under weak regularization.

Across all three settings, L1 regularization on Waterbirds follows a similar trend to Colored MNIST: weak L1 allows the spurious feature to dominate broadly across layers, moderate L1 partially limits spurious feature spread to deeper layers while preserving core feature learning, and strong L1 flattens the representational structure across all layers without a clear separation between spurious and core features. However, the effects are less pronounced than in Colored MNIST, suggesting that the Waterbirds task is inherently harder to disentangle through regularization alone, given the greater visual complexity of the background feature relative to digit color.

A.3. L2 Regularization

The Waterbirds results under L2 regularization shown in Figure 9 differ from Colored MNIST in that the spurious and core features are less cleanly separated across layers. At $\lambda = 1e-5$ and $\lambda = 1e-3$, the water (spurious) probe is highly accurate across all layers, reaching 91–93% uniformly from layer1 through avgpool. Unlike Colored MNIST, where color was concentrated in early layers, the background feature here is readable at every depth. The bird (core) probe improves gradually with both depth and training, climbing from the low 80s at relu to around 91–92% at avgpool by the final epoch. The two weaker regularization settings produce nearly identical heatmaps, suggesting that L2 at these strengths has little effect on the internal feature structure for this dataset.

At $\lambda = 0.1$, the water probe drops substantially in the early and middle layers, with relu through layer2 stuck near 74% (close to the majority class baseline of the dataset). Water only becomes readable in layer4 and avgpool, where it reaches around 88%. The bird probe is also weakened: deeper layers plateau around 85–86%, well below the 91–92% seen at lower regularization strengths. As with Colored MNIST, aggressive L2 suppresses both features, but the effect is less dramatic because the overall accuracy range is narrower and neither feature achieves the near-perfect probe accuracies seen with colored digits.

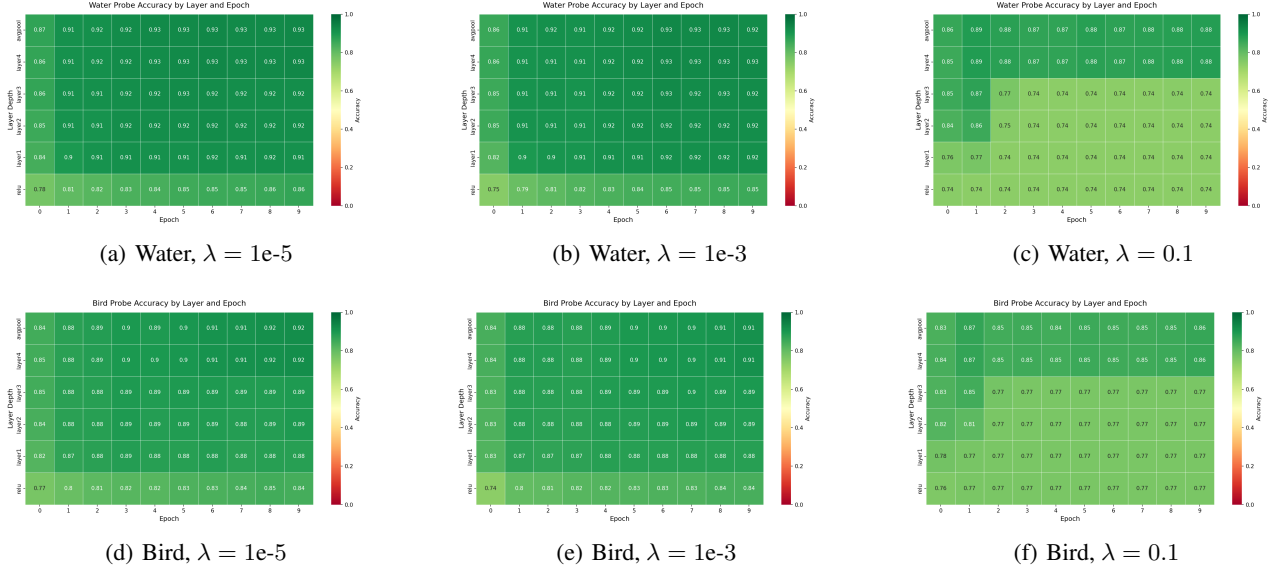


Figure 9. L2 regularization probe accuracies on Waterbirds. Top row: water (spurious) probe. Bottom row: bird (core) probe. Regularization strength increases left to right.

A.4. Dropout

The Waterbirds results under dropout regularization shown in Figure 10 closely mirror the L2 trends, with little variation in feature structure across dropout strengths. At all three settings ($p = 0.1, 0.3$, and 0.5), the water (spurious) probe is highly accurate across all layers, reaching 91–93% uniformly from layer1 through avgpool by the final epoch, with relu lagging slightly at 85–87%. Unlike Colored MNIST, where increasing dropout progressively suppressed color in deeper layers, the background feature here remains readable at every depth regardless of dropout strength. The bird (core) probe also shows minimal variation across settings, climbing gradually from the mid 70s at relu to around 89–90% at avgpool by the final epoch. The three dropout strengths produce nearly identical heatmaps, suggesting that dropout has little effect on the internal feature structure for this dataset.

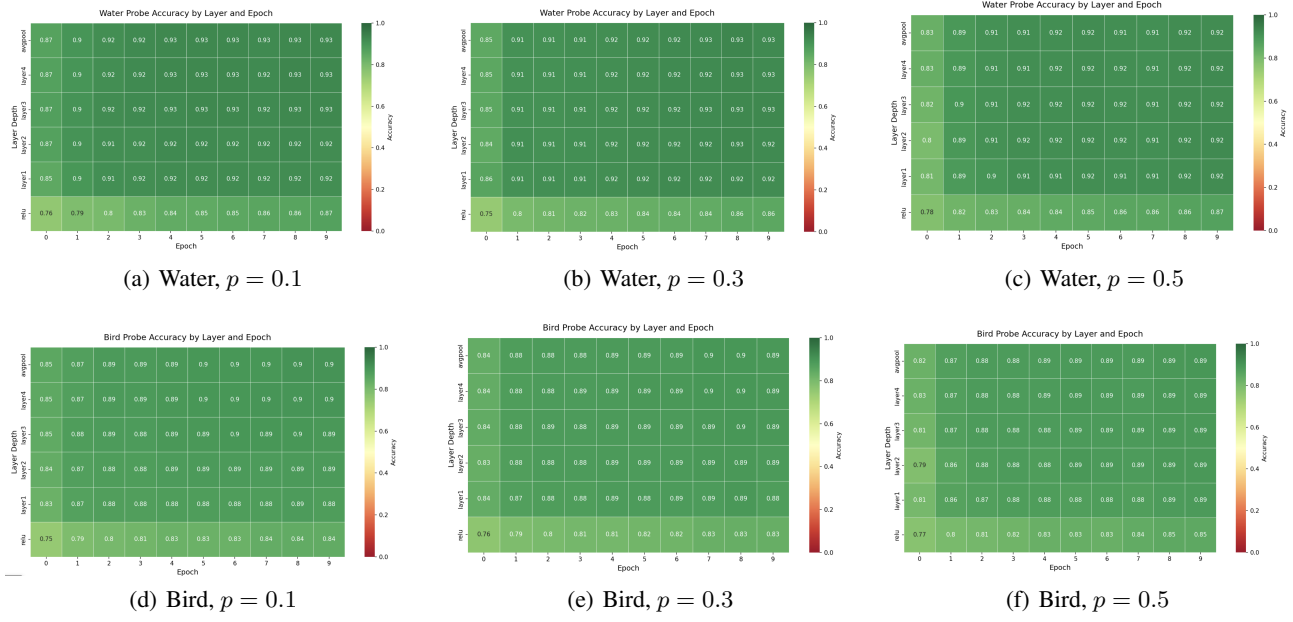


Figure 10. Dropout regularization probe accuracies on Waterbirds. Top row: water (spurious) probe. Bottom row: bird (core) probe. Dropout strength increases left to right.